

# The Mediterranean as an Earth-Scale Geophysical Sensor: Polarization Sensing on Subsea Fiber Infrastructure

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**Abstract** We demonstrate that commercial coherent submarine telecommunication links can operate as ultra-low-frequency environmental sensors by monitoring polarization telemetry from DSP receivers. Multiple Mediterranean cables detected both Rayleigh waves and Earth-scale precursor anomalies preceding the 29 July 2025  $M_w$  8.8 Kamchatka earthquake at distances of  $\sim 9000$  km. ©2026 The Author(s)

## Introduction

Modern long-haul submarine optical communication systems rely on coherent detection and digital signal processing (DSP) to dynamically compensate polarization transformations induced during fiber transmission [1]. These transformations are continuously estimated through the Jones matrix [2] retrieved in real time by field-deployed coherent receivers.

Because the Jones matrix is sensitive to perturbations acting on the optical fiber, including pressure-induced birefringence variations [3], it naturally encodes information about environmental disturbances affecting the cable along its entire submerged path. Importantly, this telemetry is generated within application-specific integrated circuits (ASICs) during normal network operation and can be extracted without modifying the wet plant or interrupting traffic [3–5].

Existing sensing approaches based on interferometric optical phase [6–11] become increasingly limited at millihertz frequencies due to laser phase noise [12], preventing reliable monitoring of ultra-low-frequency geophysical processes over long submarine spans.

In this work, we demonstrate that polarization telemetry extracted from coherent receivers enables deployed Mediterranean submarine links to operate as distributed environmental sensors with sensitivity extending into the sub-millihertz band. Using this approach, multiple commercial cables detected both the Rayleigh-wave signature and distinct precursor anomalies associated with the  $M_w$  8.8 Kamchatka earthquake of 29 July 2025, despite an epicentral distance of approximately 9000 km [13–15].

## Polarization-Based Environmental Sensing

The sensing method relies on extracting the three-dimensional polarization-rotation vector from the Jones matrix estimated by the coherent-receiver DSP [14]. After compensating

the quasi-static drift through a running-average reference, the residual rotation vector encodes time-varying perturbations acting on the fiber above the corresponding cutoff frequency.

Unlike interferometric phase-based techniques, polarization measurements are intrinsically immune to laser phase noise, allowing the detection of fractional variations in the differential optical path length between the two eigenpolarizations as small as  $5 \times 10^{-17}$  in the millihertz band [3].

In this regime, spatially coherent seafloor acceleration produces hydrostatic-pressure perturbations that propagate through the seawater column and mechanically couple to the cable. The resulting birefringence modulation induces measurable changes in the polarization-rotation vector, effectively transforming the surrounding seawater into an extended sensing medium.

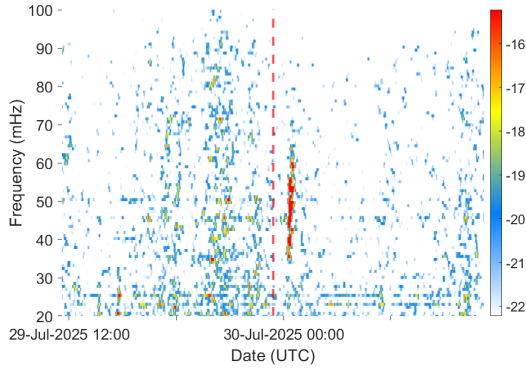
Submarine links therefore respond preferentially to large-scale coherent forcing rather than local pointwise deformation, enabling the detection of ocean-mediated geophysical dynamics that are inaccessible to land-based seismometers.

## Experimental Results

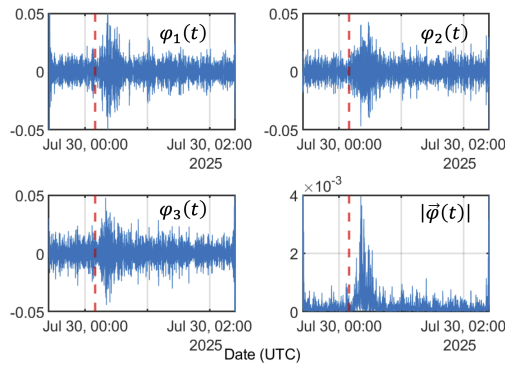
Polarization telemetry was acquired from several Mediterranean submarine links at a sampling period of 0.5 s and analyzed in the millihertz and sub-millihertz frequency ranges.

A clear Rayleigh-wave train associated with the Kamchatka earthquake was detected on the Catania–Haifa and Catania–Tel Aviv links approximately 45 min after the event, corresponding to an apparent propagation velocity of  $\sim 3.3 \text{ km s}^{-1}$  (Figs. 1 and 2).

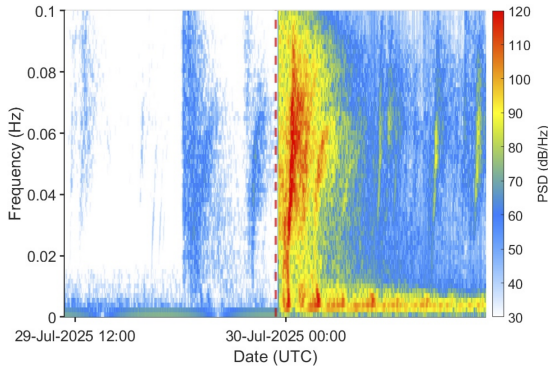
Comparison with the vertical velocity (HHZ) component recorded by the broadband seismic station GIGS at the Gran Sasso National Laboratory (Nanometrics Trillium 360 s, upgraded on 29-05-2024 from the 240 s instrument described in [16]; see <https://terremoti.ingv.it/en/instruments/station/GIGS>) shows consistent features (Figs. 3 and 4), including the small



**Fig. 1:** Spectrogram of the rotation vector of the Jones matrix obtained at the Catania terminal of the Catania–Haifa link. The time axis spans 24 hours centered on the earthquake, marked with a vertical dashed line.



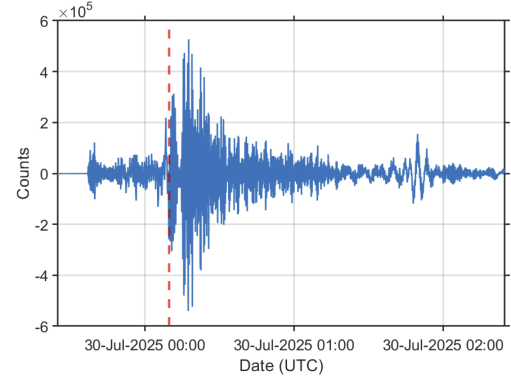
**Fig. 2:** Traces of the three components of the rotation vector  $\vec{\varphi}(t)$  and its modulus obtained from the Catania terminal of the Catania–Haifa link, filtered between 30 and 70 mHz. The time axis extends to 3 hours after the earthquake. The vertical red dashed line marks 45 minutes after the event.



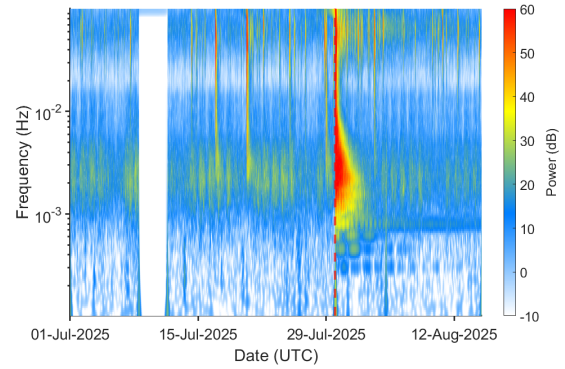
**Fig. 3:** Spectrogram of the HHZ (vertical) component recorded at the GIGS station. The time axis spans 24 hours centered on the earthquake, indicated by a vertical dashed line.

anomalous dispersion (positive slope in the time–frequency plane) of the Rayleigh-wave train.

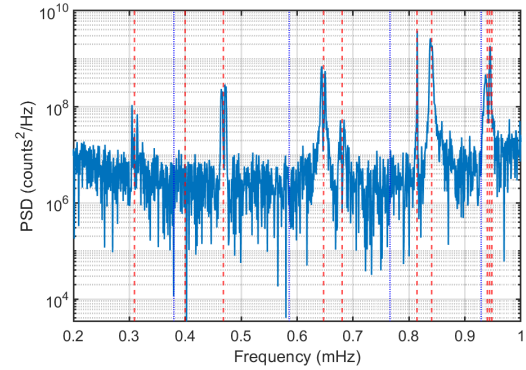
Figure 5 shows the periodogram (continuous wavelet transform with Morse wavelet) of the GIGS HHZ velocity records from 26 June to 14 August 2025. The dashed line marks the Kamchatka earthquake (29 July 2025, 23:24:52). Several low-frequency Earth normal modes of the spheroidal class  ${}_lS_n$  appear clearly, displaying temporal modulations due to the beating among their different  $m$  components. Figure 6 shows the corresponding spectrum from the earthquake time to



**Fig. 4:** Temporal traces of the HHZ velocity component recorded at GIGS over the same interval shown in Fig. 2. The vertical dashed line marks 45 minutes after the earthquake.



**Fig. 5:** Scalogram of the HHZ velocity component at GIGS from 1 June to 14 August 2025.

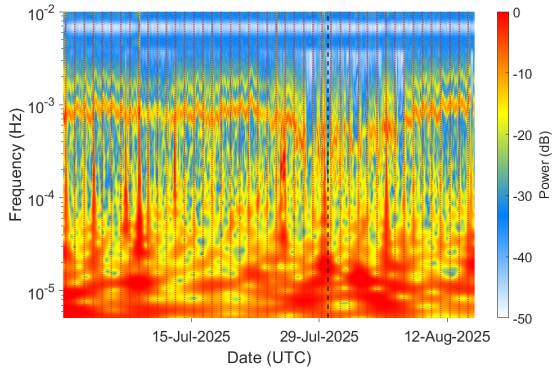


**Fig. 6:** Spectrum of the HHZ velocity component from the earthquake time to 14 August 2025. Red vertical lines indicate the frequencies of the first spheroidal Earth normal modes. Blue dotted lines refer to the frequencies of the first toroidal modes, which do not affect the vertical velocity component.

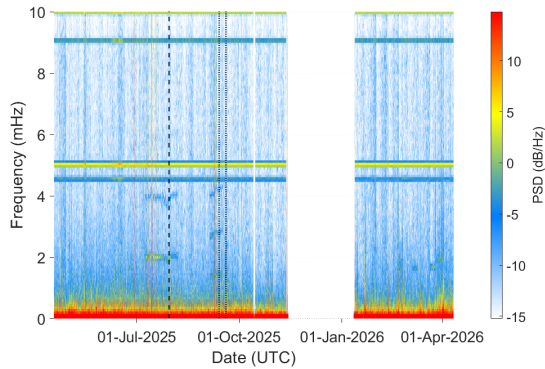
15 August 2025, with vertical red lines indicating the frequencies of the first spheroidal modes.

More significantly, narrowband resonances in the 0.3–1 mHz range and broadband impulsive activity around 0.5 mHz were observed on multiple links weeks to months prior to the mainshock [15]. Figure 7 shows the periodogram from the Roma receiver of the Roma–Golfo Aranci link over the same interval as Fig. 5. A triple resonance at multiples of 1 mHz appears prior to the earthquake, decreases in frequency, breaks shortly before the event, and reappears once the excitation of the Earth normal modes (Fig. 5) subsides.

Figure 8 shows the spectrogram from the Cha-



**Fig. 7:** Scalogram from the Roma receiver of the Roma–Golfo Aranci link from 1 July to 14 August 2025. Dotted red lines mark 07:00 UTC each day

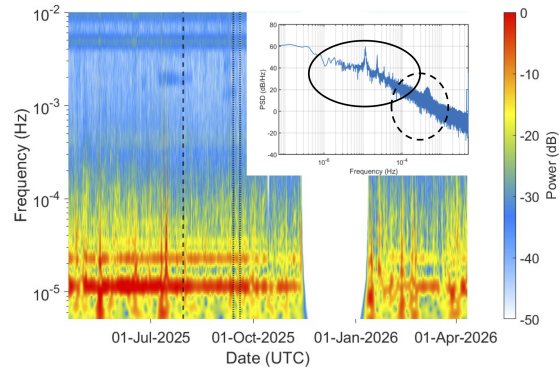


**Fig. 8:** Spectrogram from the Chania receiver of the Chania–Athens link from 18 April 2025 to 10 April 2026.

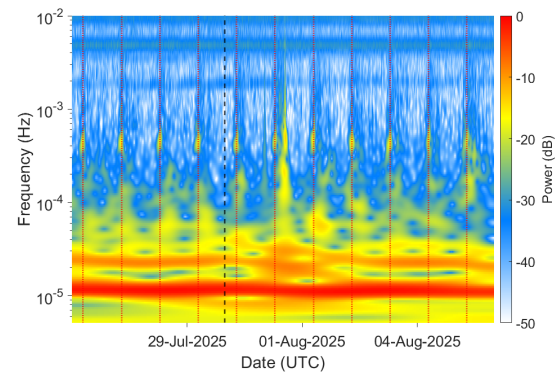
nia receiver of the Chania–Athens link, revealing a triple resonance at multiples of 2 mHz that emerges about twenty days before the earthquake, reaches a minimum at the earthquake time (dashed line), and disappears roughly ten days afterward. No comparable resonances are observed over nearly one year of measurements, except for another triple resonance coincident with the two strongest aftershocks (dotted lines). These features likely arise from shallow-water seiches near the cable. We speculate that the seiches are excited by abrupt variations in seawater pressure generated by impulsive seafloor accelerations [15].

In the Chania–Athens link, these excitations appear in the spectrogram as enhancements at the diurnal and semidiurnal frequencies, together with the excitation of the  ${}_0S_3$  Earth normal mode. The diurnal and semidiurnal features form persistent narrow lines in the scalogram of Fig. 9, which spans nearly one year of observations from 18 April 2025 to 10 April 2026. Their impulsive origin is more evident in the spectrum shown in the inset of the same figure, where multiple lines at integer multiples of the diurnal frequency are visible. The impulsive excitation of the  ${}_0S_3$  mode occurring daily at 07:00 UTC is clearly seen in Fig. 10, which zooms into the interval from 26 July 2025 to midnight on 5 August 2025.

Land-based GIGS records show no continuous Earth-normal-mode excitation prior to the earth-



**Fig. 9:** Scalogram for the same period as Fig. 8. The inset shows the spectrum of the rotation vector from 18 April 2025 to 30 September 2025. Peaks associated with the daily sequence of baseband pulses are highlighted with a solid circle, while pulses near the  ${}_0S_3$  Earth normal mode are marked with a dotted circle.



**Fig. 10:** Zoom of Fig. 9 from 26 July to 5 August 2025. Dotted red lines mark 07:00 UTC each day.

quake. This is expected because cables are sensitive to pressure modulations, that is, to seafloor acceleration, whereas broadband seismometers sense ground velocity. Impulsive accelerations that strongly affect cables may not integrate into detectable velocity variations. Further, optical-fiber links benefit from coherent enhancement of spatially coherent forcing, whereas seismometers respond only locally. This highlights the complementarity between distributed cable sensing and pointwise terrestrial instruments.

## Conclusions

We have shown that commercial submarine optical-telecommunication links can operate as large-scale environmental sensors by exploiting polarization telemetry already available from coherent DSP receivers.

The ability to detect coherent seawater-pressure perturbations associated with distant seismic activity over continental scales demonstrates that existing submarine infrastructure can provide continuous monitoring of ultra-low-frequency geophysical processes without dedicated instrumentation.

This approach opens opportunities for real-time observation of Earth-scale dynamics and may enable future early-warning systems leveraging existing communication networks.

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